STATISTICS

Regression and Correlation Analysis

PART 2: Estimation and Hypothesis for Regression Parameters

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Point Estimation of Regression Parameters

When we select a sample and then obtain a regression line

$$\hat{Y} = b_0 + b_1 X,$$

the following estimations can be made:

$$\left.\begin{array}{l} \beta_0 = b_0 \\ \beta_1 = b_1 \\ \rho = r \\ Y_k = \hat{Y}_k \text{ for a given } x_k \\ \bar{Y}_k = \hat{Y}_k \text{ for a given } x_k \end{array}\right\} \text{point estimations}$$

The above regression parameters can also be estimated by a confidence interval and/or be tested by hypothesis testing.

Correlation Coefficient ρ

For the correlation coefficient of the population ρ and its estimator r, the variable

$$t_{n-2} = \frac{r - \rho}{S_r}$$

has t distribution with freedom degree n-2. Here n is the size of the sample and

$$S_r = \sqrt{\frac{1 - r^2}{n - 2}}$$

is the standard deviation of the estimator r. S_r is also known as the standard error of the correlation coefficient.

Correlation Coefficient ρ

Confidence Interval for ρ:

The confidence interval with significance level α can be derived as:

$$P\left[-t_{\alpha/2,n-2} \le \frac{r-\rho}{S_R} \le t_{\alpha/2,n-2}\right] = 1-\alpha$$

$$\Downarrow$$

$$P\left[r - t_{\alpha/2,n-2}S_R \le \rho \le r + t_{\alpha/2,n-2}S_R\right] = 1-\alpha$$

$$\Downarrow$$

$$\left[r - t_{\alpha/2,n-2}S_R, \quad r + t_{\alpha/2,n-2}S_R\right]$$

Correlation Coefficient ρ

Hypothesis Testing for ρ:

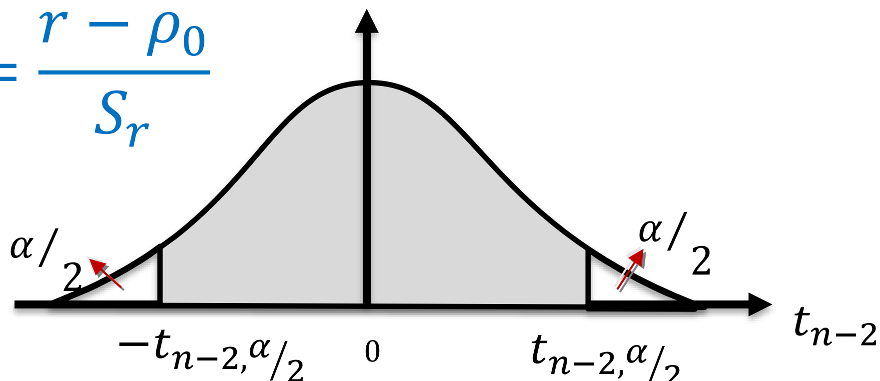
While it is possible to test several one-tailed and two-tailed hypotheses, the most commonly used hypotheses are:

$$H_0 : \rho = 0\ (\rho_0) \quad \text{(There is no correlation.)}$$

$$H_a : \rho \neq 0\ (\rho_0) \quad \text{(There is a correlation.)}$$

The test statistic is: $t_0 = \dfrac{r - \rho_0}{S_r}$

The decision rule:



Reject H_0, if $-t_{n-2,\alpha/2} > t_0$ or $t_0 > t_{n-2,\alpha/2}$

If we reject H_0, it means that there is a significant relation between *X* and *Y*. If there is no logical relation between variables, the correlation is said to be fake (correlation does not imply causation).

Example: In a duration of one hour, the number of customers coming to a shop and their spending are given below.

The number of customers	25	20	50	35	40
Spendings (x 100 TL)	12.5	10.4	25.3	20.2	24.1

If we test the correlation coefficient at 5% importance level, test hypotheses are

$$H_0 : \rho = 0$$

$$H_1 : \rho \neq 0$$

$$t_0 = \frac{r - \rho_0}{S_r}$$

$$n - 2 = 5 - 2 = 3$$

$$\alpha/2 = 0.025$$

$$t_{n-2,\alpha/2} = 3.182$$

If we assume that the independent variable *X* is the number of customers and the dependent variable *Y* is the amount of spending, the correlation coefficient is;

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{\left[n\sum X^2 - (\sum X)^2\right]\left[n\sum Y^2 - (\sum Y)^2\right]}}$$

	X	Y	XY	X^2	Y^2
	25	12.5	312.5	625	156.2
	20	10.4	208	400	108.1
	50	25.3	1265	2500	640.09
	35	20.2	707	1225	408.04
	40	24.1	964	1600	580.81
Total	170	92.5	3456.5	6350	1893.3

$$r = \frac{5(3456.5) - 170(92.5)}{\sqrt{\left[5(6350 - 170^2\right]\left[5(1893.3) - (92.5)^2\right]}} = 0.9669$$

$$t_0 = \frac{r - \rho_0}{\sqrt{\frac{1 - r^2}{n - 2}}} = \frac{0.9669 - 0}{\sqrt{\frac{1 - (0.9669)^2}{5 - 2}}} = 6.5635$$

$t_{n-2,\alpha/2} = 3.182$

$t_0 = 6.5635$

$\Rightarrow$ $t_0 > t_{n-2,\alpha/2}$

$\Downarrow$

We reject hypothesis H_0 for the 5% importance level. There is a correlation between variables.

Population Parameters β_0 and β_1

The means and variances of the statistics b_0 and b_1 are:

$$E(b_0)=\beta_0 \qquad \mathrm{Var}(b_0) = \frac{\sum X^2}{n\sum X^2 - (\sum X)^2}\sigma^2$$

$$E(b_1)=\beta_1 \qquad \mathrm{Var}(b_1) = \frac{1}{[\sum X^2 - (\sum X)^2/n]}\sigma^2$$

where σ^2 is the variance of Y.

If the statistics b_0 and b_1 are distributed normally, the variables:

$$z = \frac{b_0 - \beta_0}{\sqrt{Var(b_0)}} \qquad z = \frac{b_1 - \beta_1}{\sqrt{Var(b_1)}}$$

have normal distributions.

If we do not know the population parameter σ^2, instead of it we can use its estimater

$$S^2 = \frac{\sum\left(Y - \hat{Y}\right)^2}{n-2} = \frac{\sum e^2}{n-2}$$

which is called the variance of estimation (estimator). Here, S is the standard error of estimation (estimator).

In this case, the variables distributed normally are converted into:

$$t_{n-2} = \frac{\mathrm{b}_0 - \beta_0}{\sqrt{Var(\mathrm{b}_0)}} \qquad t_{n-2} = \frac{\mathrm{b}_1 - \beta_1}{\sqrt{Var(\mathrm{b}_1)}}$$

where

$$\mathrm{Var}(b_0) = \frac{\sum X^2}{n\sum X^2 - (\sum X)^2} S^2 \qquad \mathrm{Var}(b_1) = \frac{1}{[\sum X^2 - (\sum X)^2/n]} S^2$$

Population Parameters β_0 and β_1

Confidence Interval for β_0 and β_1 :

The confidence interval with significance level α can be derived as:

$$P\left[-t_{\alpha/2,n-2} \le \frac{b_0-\beta_0}{\sqrt{\mathrm{Var}(b_0)}} \le t_{\alpha/2,n-2}\right] = 1-\alpha$$

$$\Downarrow$$

$$P\left[b_0 - t_{\alpha/2,n-2}\sqrt{\mathrm{Var}(b_0)} \le \beta_0 \le b_0 + t_{\alpha/2,n-2}\sqrt{\mathrm{Var}(b_0)}\right] = 1-\alpha$$

$$\Downarrow$$

$$\left[b_0 - t_{\alpha/2,n-2}\sqrt{\mathrm{Var}(b_0)}, \quad b_0 + t_{\alpha/2,n-2}\sqrt{\mathrm{Var}(b_0)}\right]$$

Similarly for the other parameter:

$$\left[b_1 - t_{\alpha/2,n-2}\sqrt{\mathrm{Var}(b_1)}, \quad b_1 + t_{\alpha/2,n-2}\sqrt{\mathrm{Var}(b_1)}\right]$$

Population Parameters β_0 and β_1

Hypothesis Testing for β_0 and β_1 :

The most commonly used hypotheses for β_0 and β_1 are:

$H_0 : \beta_0 = a$
$H_a : \beta_0 \neq a$ $\Rightarrow$ test statistic: $t_0 = \dfrac{b_0 - \beta_0}{\sqrt{\mathrm{Var}(b_0)}}$, $(\beta_0 = a)$

If $-t_{n-2,\alpha/2} > t_0$ or $t_0 > t_{n-2,\alpha/2}$, we reject H_0.

$H_0 : \beta_1 = b$
$H_a : \beta_1 \neq b$ $\Rightarrow$ test statistic: $t_1 = \dfrac{b_1 - \beta_1}{\sqrt{\mathrm{Var}(b_1)}}$, $(\beta_1 = b)$

If $-t_{n-2,\alpha/2} > t_1$ or $t_1 > t_{n-2,\alpha/2}$, we reject H_0.

$H_0 : \beta_1 = 0$ means that β_1 is not significant statistically or there is no linear relationship between *X* and *Y*. (*Y* may be dependent on *X*, but not linearly.)

$H_0 : \beta_0 = 0$ means that β_0 is not significant statistically.

The Estimation of Y_k for a Specific Value of X_k

One of the most important purposes of regression analysis is to estimate the value of Y_k for a value of X_k.

$$\widehat{Y_k} = b_0 + b_1 X_k$$

Here, $\hat{Y}_k$ is an estimator of the population parameter Y_k. The variable

$$t_{n-2} = \frac{\hat{Y}_k - Y_k}{S_{\hat{Y}_k}}$$

t distribution with freedom degree n-2. Here n is the size of the sample and the standard error of the estimation is

$$s_{\widehat{Y_k}} = \sqrt{1 + \frac{1}{n} + \frac{(X_k - \bar{X})^2}{\sum X^2 - n\bar{X}^2}}\, S$$

The Estimation of Y_k for a Specific Value of X_k

Note that if we interested in the mean of Y_k for a specific value of X_k in the population, the standard error of the estimation becomes

$$S_{\widehat{\mu_{Y_k}}} = \sqrt{\frac{1}{n} + \frac{(X_k - \bar{X})^2}{\sum X^2 - n\bar{X}^2}}\, S$$

Y_k for a given x_k $\Rightarrow$ $$S_{\widehat{Y_k}} = \sqrt{1 + \frac{1}{n} + \frac{(X_k - \bar{X})^2}{\sum X^2 - n\bar{X}^2}}\, S$$

For example, the maintenance spending of a specific truck whose age is 4 years…

$\bar{Y}_k$ for a given x_k $\Rightarrow$ $$S_{\widehat{\mu_{Y_k}}} = \sqrt{\frac{1}{n} + \frac{(X_k - \bar{X})^2}{\sum X^2 - n\bar{X}^2}}\, S$$

For example, the mean of the maintenance spending of trucks whose ages are 4 years…

The Estimation of Y_k for a Specific Value of X_k

Confidence Interval for Y_k:

The confidence interval with significance level α can be derived as:

$$P\left[-t_{\alpha/2,n-2} \le \frac{\widehat{Y_k} - Y_k}{S_{\widehat{Y_k}}} \le t_{\alpha/2,n-2}\right] = 1 - \alpha$$

$$\Downarrow$$

$$P\left[\widehat{Y_k} - t_{\alpha/2,n-2} S_{\widehat{Y_k}} \le Y_k \le \widehat{Y_k} + t_{\alpha/2,n-2} S_{\widehat{Y_k}}\right] = 1 - \alpha$$

$$\Downarrow$$

$$\left[\widehat{Y_k} - t_{\alpha/2,n-2} S_{\widehat{Y_k}}, \quad \widehat{Y_k} + t_{\alpha/2,n-2} S_{\widehat{Y_k}}\right]$$

ANOVA: Population Parameters β_0 and β_1

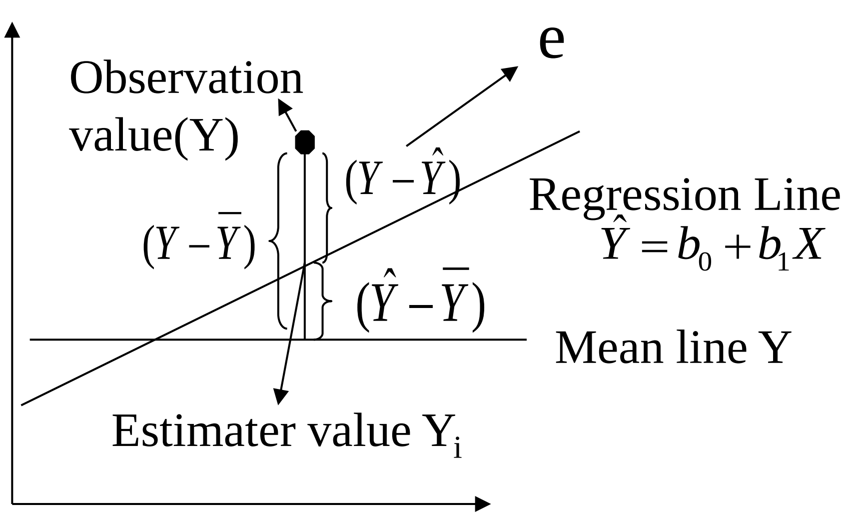
$$\sum(Y-\bar{Y})^2 = \sum(\hat{Y}-\bar{Y})^2 + \sum(Y-\hat{Y})^2$$

Sum of Total Squares (SST)

Sum of Regression Squares (SSR)

Sum of Error Squares (SSE)

SST = SSR +SSE



Thus, the coefficient of determination:

$$R^2 = \frac{SRS}{STS}$$

ANOVA Table for β_0 and β_1

The population parameters β_0 and β_1 can be tested separately as given before. However, they also can be tested by using a model containing both parameters at the same time. Accordingly, the hypothesis test can be applied using the table given below:

Variance Source	Square Sum	Degree of Freedom	Average Squares	The Proportion F
Regression	SSR	1	SRS/1	
Error	SSE	n-2	SES/(n-2)	$F_{1,n-2,\alpha} = \frac{SRS}{\frac{SES}{n-2}}$
Total	SST	n-1		

$H_0 : \beta_0 = \beta_1 = 0$

$H_1 : \beta_0 \neq 0$ or $\beta_1 \neq 0$

Test Statistic: $F_0 = \frac{SSR}{SSE/_{n-2}}$

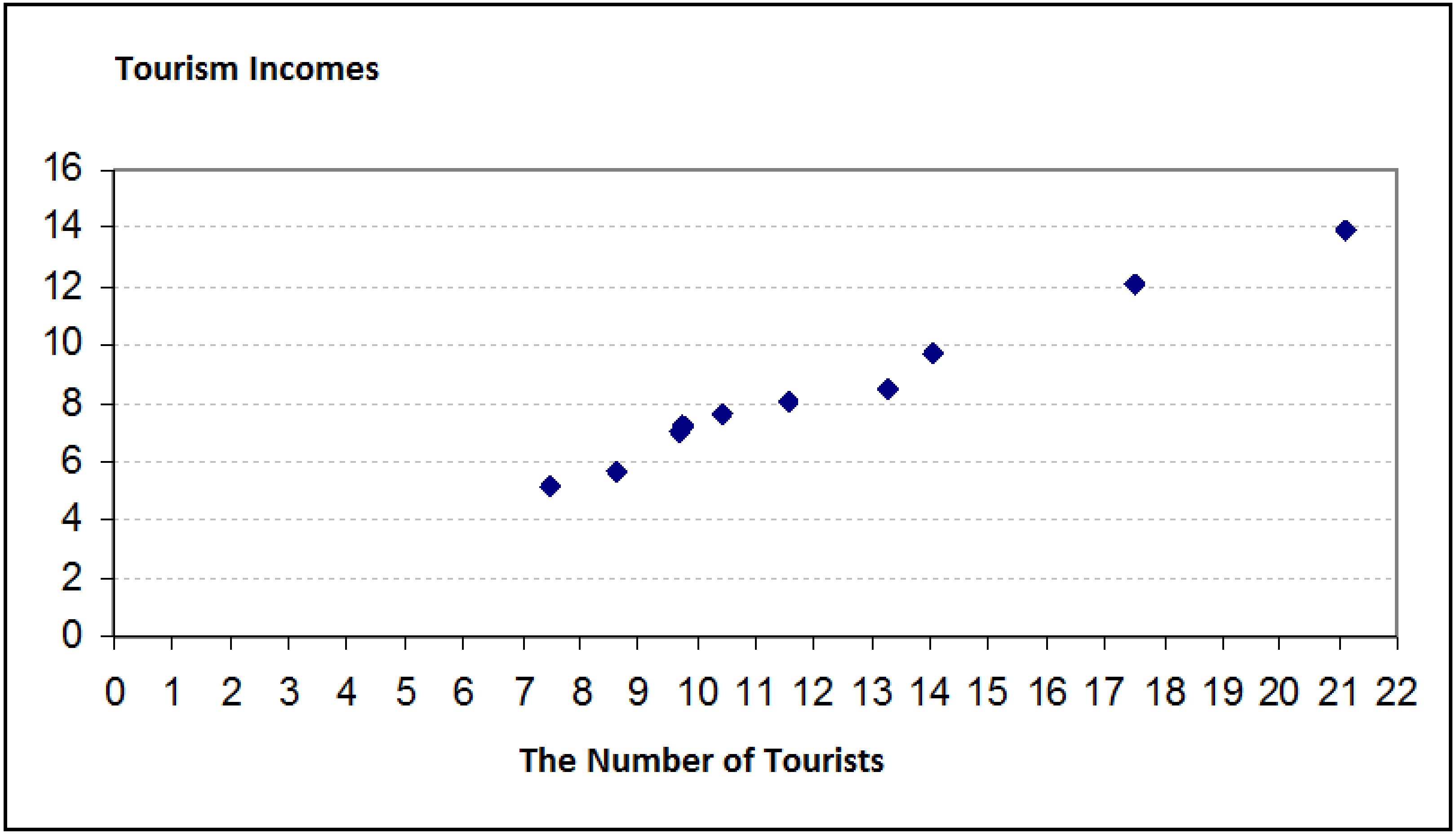
If $F_0 > F_{1,n-2,\alpha}$, then H_0 is rejected.

Example: In the years 1996-2005, the tourism incomes of Turkey and the number of tourists who visited Turkey are given below.

Years	Tourism Incomes	The Number of Tourists
1996	5.650	8.614
1997	7.008	9.689
1998	7.177	9.752
1999	5.193	7.464
2000	7.636	10.412
2001	8.090	11.569
2002	8.481	13.247
2003	9.677	14.030
2004	12.125	17.517
2005	13.929	21.122

1. Draw the scatter diagram.
2. Find the estimated regression line.
3. Compute the standard error and variance of the estimator for the estimation of β_0 and β_1.
4. Compute the variances of the coefficients b_0 and b_1
5. Construct the confidence intervals for β_0 and β_1 (α=0.05).
6. Test the hypothesis for $H_0 : \beta_0 = 0$ and $H_a : \beta_0 \neq 0$ (α=0.05).
7. Test the hypothesis for $H_0 : \beta_1 = 0$ and $H_a : \beta_1 \neq 0$ (α=0.05).
8. Compute SSR, SSE and SST.
9. Compute the determination coefficient and interpret it.
10. Compute the correlation coefficient and interpret it.
11. Test the significance of the regression through the table of variance analysis.
12. Construct the confidence interval of estimation Y_k (α=0.05) for $X_k = 8.614$.

1. The Scatter Diagram



2. The Regression Line

Y	X	Y*X	X^2
5.650	8.614	48.6691	74.201
7.008	9.689	67.9005	93.8767
7.177	9.752	69.9901	95.1015
5.193	7.464	38.7605	55.7113
7.636	10.412	79.5060	108.4097
8.090	11.569	93.5932	133.8418
8.481	13.247	112.3478	175.4830
9.677	14.030	135.7683	196.8409
12.125	17.517	212.3936	306.8452
13.929	21.122	294.2083	446.1388
$\Sigma Y=84.966$	$\Sigma X=123.416$	$\Sigma YX=1153.138$	$\Sigma X^2=1686.4501$

2. The Regression Line

$$\Sigma Y = b_0.n + b_1.\Sigma X$$
$$\Sigma YX = b_0.\Sigma X + b_1.\Sigma X2$$

$$84.96 = b_0.10 + b_1.\ 123.4$$
$$1153.13 = b_0.123.4 + b_1.\ 1686.4$$

$b_0=0.597$ $b_1=0.640$

$$\hat{Y} = 0.597 + 0.640X$$

When the number of tourists increases, the amount of tourism income increases.

3. The Standard Error and Variance of the Estimators for the Computation β_0 and β_1

$$s = \sqrt{\frac{\sum e^2}{n-2}} \qquad s^2 = \frac{\sum e^2}{n-2}$$

Y	Y^2	$\hat{Y} = 0.597 + 0.640X$	$e = Y - \hat{Y}$	e^2
5.65	31.92	0.597 + 0.640(**8.614**) = 6.1099	-0.460	0.2115
7.008	49.11	0.597 + 0.640(**9.689**) = 6.7979	0.210	0.0441
7.177	51.51	0.597 + 0.640(**9.752**) = 6.8382	0.339	0.1147
5.193	26.96	0.597 + 0.640(**7.464**) = 5.3739	-0.181	0.0327
7.636	58.31	0.597 + 0.640(**10.412**) = 7.2606	0.375	0.1408
8.09	65.45	0.597 + 0.640(**11.569**) = 8.0011	0.089	0.0078
8.481	71.93	0.597 + 0.640(**13.247**) = 9.0750	-0.594	0.3529
9.677	93.65	0.597 + 0.640(**14.030**) = 9.5762	0.101	0.0101
12.125	147.02	0.597 + 0.640(**17.517**) = 11.8078	0.317	0.1005
13.929	194.02	0.597 + 0.640(**21.122**) = 14.1150	-0.186	0.0346
	$\Sigma Y^2 = 789.8721$	$\sum \hat{Y} = 84.966$	0.010	$\Sigma e^2 = 1.0501$

$$s = \sqrt{\frac{\sum e^2}{n-k}} = \sqrt{\frac{1.0501}{10-2}} = 0.362$$

$$s^2 = (0.362)^2 = 0.131$$

4. The Variances of the Coefficients b_0 and b_1

$$\text{Var}(b_0) = \frac{\sum X^2}{n\sum X^2 - (\sum X)^2} S^2 \quad = 0.134$$

$$\text{Var}(b_1) = \frac{1}{[\sum X^2 - (\sum X)^2/n]} S^2 = 0.00078$$

The standard error of the coefficients b_0 and b_1

$$\sqrt{\text{Var}(b_0)} = \sqrt{0.134} = 0.367$$

$$\hat{Y} = 0.597 + 0.640X$$

$$\sqrt{\text{Var}(b_1)} = \sqrt{0.00078} = 0.028$$

5.The confidence intervals for β_0 and β_1 (α=0.05)

$$\left[b_0 - t_{\alpha/2, n-2}\sqrt{\mathrm{Var}(b_0)}, \quad b_0 + t_{\alpha/2, n-2}\sqrt{\mathrm{Var}(b_0)} \right]$$

[0.597- 2.306 (0.367), 0.597+ 2.306 (0.367)]

[0.597 - 0.8463, 0.597 + 0.8463]

$0.2493 \leq \beta_0 \leq 1.4433$

$$\left[b_1 - t_{\alpha/2, n-2}\sqrt{\mathrm{Var}(b_1)}, \quad b_1 + t_{\alpha/2, n-2}\sqrt{\mathrm{Var}(b_1)} \right]$$

[0.640 - 2.306 (0.028) , 0.640 + 2.306 (0.028)]

[0.640 - 0.0645, 0.640 + 0.0645

$0.5755 \leq \beta_1 \leq 0.7045$

6. Hypothesis Testing for β_0 (α=0.05)

$H_0 : \beta_0 = 0$ $\quad$ $t_{0.05/2\ ,\ 8} = 2.306$

$H_a : \beta_0 \neq 0$

$$t_0 = \frac{b_0 - \beta_0}{\sqrt{\mathrm{Var}(b_0)}} = \frac{0.597 - 0}{0.367} = 1.626$$

$t_0 = 1.626 < t_{0.05/2\ ,\ 8} = 2.306$

Accept H_0.

β_0 is not statistically significant.

7. Hypothesis Testing for β_1 (α=0.05)

$H_0 : \beta_1 = 0$

$H_a : \beta_1 \neq 0$

$t_{0.05/2\ ,\ 8} = 2.306$

$$t_1 = \frac{b_1 - \beta_1}{\sqrt{\mathrm{Var}(b_1)}} = \frac{0.640 - 0}{0.028} = 22.85$$

$t_0 = 22.85 > t_{0.05/2\ ,\ 8} = 2.306$

Reject H_0.

β_1 is statistically significant. There is a linear relationship between *X* and *Y*.

8. SST, SSR and SSE

$$\hat{Y} = 0.597 + 0.640X$$

Sum of Total Squares (SST) $\sum(Y-\bar{Y})^2 = \sum Y^2 - (\sum Y)^2 / n$

Sum of Regression Squares (SSR) $\sum(\hat{Y}-\bar{Y})^2 = b_1^2\left[\sum X^2 - (\sum X)^2 / n\right]$

Sum of Error Squares (SSE) $\sum(Y-\hat{Y})^2 = SST - SSR$

$\Sigma X^2=1686.4501$ $\Sigma X=123.416$ $\Sigma Y=84.966$ $\Sigma Y^2=789.8721$

SST ➡ $\sum(Y-\bar{Y})^2 = \sum Y^2 - (\sum Y)^2 / n = 789.8721 - 7219.229/10 = 67.9499$

SSR ➡ $b_1^2\left[\sum X^2 - \frac{(\sum X)^2}{n}\right] = (0.640)^2\left[1686.4501 - \frac{(123.416)^2}{10}\right] = 66.8873$

SSE ➡ $SSE = \sum e^2 = 1.0501 \cong 67.9499 - 66.8873$

9. Coefficient Determination R^2 :

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2} = \frac{66.8873}{67.9499} \cong 0.9844$$

Comment: This result shows that 98.44% of the variation of tourism incomes dependent on the number of tourists. (98.44% of the variability in the data is explained by the regression model)

10. Correlation Coefficient

$$r = \sqrt{0.98.44} = 0.9922 \longrightarrow$$

$$\hat{Y} = 0.597 + 0.640X$$

There is a positive (since the slope of the regression line is positive) strong relationship between tourism incomes and the number of tourists.

11. Test the significance of the regression through the table of variance analysis.

$$H_0 : \beta_0 = \beta_1 = 0$$
$$H_1 : \beta_0 = \beta_1 \neq 0$$

	Degree of freedom	*SS*	*Average*
SSR	1	66.8873	66.8873
SSE	8	1.0501	0.1312
SST	9	67.9499	

$$SSR = (0.640)^2 \left[1686.4501 - \frac{(123.416)^2}{10} \right] = 66.8873$$

$$SSE = \sum e^2 = 1.0501$$

$$SST = \sum (Y - \bar{Y})^2 = \sum Y^2 - (\sum Y)^2 / n = 789.8721 - 7219.229/10 = 67.9499$$

$$F_0 = \frac{66.8873}{0.1312} = 509.81 \qquad F_{0.05,1,8} = 5.32$$

$F_0 > F_{0.05,1,8}$; reject H_0; at least one of the parameters (β_0 and β_1) is statistically significant.

12. The confidence interval of estimation Y_k (α=0.05) for X_k = 8.614.

$$\left[\widehat{Y_k} - t_{\alpha/2, n-2} S_{\widehat{Y_k}}, \qquad \widehat{Y_k} + t_{\alpha/2, n-2} S_{\widehat{Y_k}}\right]$$

$$S_{\widehat{Y_k}} = \sqrt{1 + \frac{1}{n} + \frac{(X_k - \bar{X})^2}{\sum X^2 - n\bar{X}^2}}\, S$$

$$X_k = 8.614$$

$$\hat{Y}_k = 6.1099$$

$$\alpha = 0.05$$

$$\alpha/2 = 0.025 \longrightarrow t_{0.025,8} = 2.306$$

$$n - 2 = 8$$

$$6.1099 \pm 2.306\sqrt{1 + \frac{1}{10} + \frac{(8.614 - 12.3416)^2}{163.2991}}\ 0.362$$

$$5.20124 \le Y_k \le 7.0180$$